

1. The _____ is a general purpose programmable I/O device designed to transfer the data from I/O to interrupt I/O.
The 8255A is a general purpose programmable I/O device designed to transfer the data from I/O to interrupt I/O
2. How many ports 8255A has?
8255A has three ports, i.e., PORT A, PORT B, and PORT C.
3. Which port can be split into two parts?
Port C can be split into two parts, i.e. PORT C lower (PC0-PC3) and PORT C upper (PC7-PC4) by the control word.
4. What are the Features of 8255A?
It consists of 3 8-bit IO ports i.e. PA, PB, and PC.
Address/data bus must be externally demux'd.
It is TTL compatible.
5. What is responsible for controlling the internal/external transfer of data/control/status word?
Read/Write Control Logic : This block is responsible for controlling the internal/external transfer of data/control/status word.
6. Which of the following uses N-MOS technology?

A.	8253
B.	8254
C.	8255
D. 8256	
7. It is a tri-state, bi-directional, 8-bit buffer, which is used to interface the 8253/54 to the system data bus.
Data Bus Buffer : It is a tri-state, bi-directional, 8-bit buffer, which is used to interface the 8253/54 to the system data bus.
8. 8253/54 can be operated in _____ Modes?
8253/54 can be operated in 6 different modes
9. Which mode can be used as a mono stable multi-vibrator for 8253/54?
Mode 1: Programmable One Shot can be used as a mono stable multi-vibrator.
10. Which mode of 8253/54 generates a strobe in response to an externally generated signal?
Mode 5 : Hardware Triggered Mode generates a strobe in response to an externally generated signal.
11. How many types of Interfacing?
Interface is the path for communication between two components. Interfacing is of two types, memory interfacing and I/O interfacing.
12. In which type of communication, the interface gets a single byte of data from the microprocessor and sends it bit by bit to the other system serially and vice-a-versa?
Serial Communication Interface : In this type of communication, the interface gets a single byte of data from the microprocessor and sends it bit by bit to the other system serially and vice-a-versa.
13. Which of the following are known as Higher Address Bus?

A.	A15	-	A8
B.	AD7	-	AD0
C.			READY
D. WR			

14. In which mode, the CPU periodically reads an internal flag of 8279 to check whether any key is pressed or not with key pressure?
In the Polled mode, the CPU periodically reads an internal flag of 8279 to check whether any key is pressed or not with key pressure.
15. What is the function of Encoded Mode?
In the encoded mode, the counter provides the binary count that is to be externally decoded to provide the scan lines for the keyboard and display.
16. Which pin is used to blank the display during digit switching?
BD : It stands for blank display. It is used to blank the display during digit switching.
17. Which mode allows 8/16 character multiplexed displays to be organized as dual 4-bit/single 8-bit display units?
Display Scan : This mode allows 8/16 character multiplexed displays to be organized as dual 4-bit/single 8-bit display units.
18. DMA stands for?
DMA stands for Direct Memory Access
19. What are the features of 8257?
A. It has four channels which can be used over four I/O devices is true.
B. Each channel has 16-bit address and 14-bit counter.
C. Each channel can transfer data up to 64kb.
D. Each channel can be programmed independently.
20. What is range of frequency for 8257?
Its frequency ranges from 250Hz to 3MHz.
21. An 8-bit microprocessor can process _____ data at a time.
An 8-bit microprocessor can process 8-bit data at a time.
22. The number of bits processed in a single instruction is known as?
Bandwidth : It is the number of bits processed in a single instruction.
23. Which is the pin efficient method of communicating between other devices?
The serial ports are considered to be the pin efficient method of communication between other devices within an embedded system.
24. Which depends the number of bits that are transferred?
The time taken for the data transmission within the system depends on the clock frequency and the number of bits that are transferred.
25. Which is the most commonly used buffer in the serial porting?
Most of the serial ports uses a FIFO buffer so that the data is not lost. The FIFO buffer is read to receive the data, that is, first in first out.
26. What does SPI stand for?
The serial parallel interface bus is a commonly used interface which involves master slave mechanism. The shift registers are worked as master and the slave devices are driven by a common clock.
27. Which allows the full duplex synchronous communication between the master and the slave?
The serial peripheral interface allows the full duplex synchronous communication between the master and the slave devices. MC68HC05 developed by Motorola uses SPI for interfacing the peripheral devices.
28. How many data lines does 256*4 have?

There are four data lines in the memory and these different organisations of memory and these different organisations of memory are apparent when upgrading memory and it also determines how many chips are needed.

29. How is the number of chips required is determined?

The minimum number of chips is determined by the number of data lines and the width of the data path from the processor. For example, MC6800 family have a 16-bit wide datapath, 16*1 devices, 4*4 or 2*8 devices are needed.

30. Where is memory address stored in a C program?

Memory model is defined by a range of memory address which is accessible to the program. For example, in the C program, the memory address is stored in the pointer.

31. Which is the term that is used to refer the order of bytes?

Endianness defines the order of bytes, that is, whether it is big endian or little endian. The former represents the higher order bits and the latter represents the lower order bits.

32. What does SIMM stand for?

SIMM is single in-line memory module is a kind of memory module, which contains random access memory used in computers of the early 1980s and 1990s.

33. Which can destroy the accuracy in the algorithms?

The delays occurring in the memory management unit can destroy the accuracy in the algorithms and in order to avoid this, the linear addressing range should be increased.

34. Which step detects the key in a 4*4 keyboard matrix about the key that is being pressed?

For detecting that whether the key is actually pressed or not, firstly this must be ensured that initially all the keys are closed. Then we need to mask the bits individually to detect that which key is pressed. Then we need to check that is the key actually pressed or not, by checking that whether the key pressed for a time more than 20 micro seconds.

35. In reading the columns of a matrix, if no key is pressed what should we get all in binary notation?

If no key is pressed, then all the keys show 1 as they are all connected to power supply.

36. If we need to operate a key of a keyboard in an interrupt mode, then it will generate what kind of interrupt?

If a key is to operate in an interrupt mode then it will generate an external hardware interrupt.

37. To identify that which key is being pressed, what we need to do?

To detect that which key is being pressed, we need to ground the pins one by one.

38. How many data lines are there in a 16*2 alphanumeric LCD?

There are eight data lines from pin no 7 to pin no 14 in an LCD.

39. Which pin of the LCD is used for adjusting its contrast?

Pin no 3 is used for controlling the contrast of the LCD.

40. For writing commands on an LCD, what is RS bit?

For writing commands on an LCD, RS pin is reset.

41. Which command of an LCD is used to shift the entire display to the right?

0x1C is used to shift the entire display to the right.

42. Which instruction is used to select the first row first column of an LCD?

0x80 is used to select the first row first column of an LCD.

43. Which pins of a microcontroller are directly connected with 8255?

RD, WR, D0-D7 all are directly connected to the 8051 for telling the chip about the control signals and also for transferring the data.

44. Find the control word for PA = out, PB = in, PCL = out, PCH = out (Mode0)?
The value that is being loaded in the control word is 10000010b for PB as an input port and all others as the output ports being operated in mode0. The hex equivalent of 10000010b $\rightarrow$ 0x82H.
45. Which pins are used to select the ports and the control register?
CS pin is an active low input pin for 8255 and it is used for selecting a chip. A0 and A1 pins are used for select ports and the control register.
46. What is the value of the control register when RESET button is set to zero?
RESET is active-high signal input into the 8255 used to clear the control register. When RESET is activated (i.e. set to high), all ports are initialized as input mode. Hence the value of the control register remains the same as it is even when the RESET button is set to zero.
47. Why MOVX instruction is being used to access the ports of the 8255?
As 8255 is connecting a microcontroller in memory mapped I/O configuration. This means that memory space used to access 8255 (i.e. 8255 is treated as external memory). MOVX instruction is used to access external memory locations.
48. What is the BSR mode in 8255?
BSR (Bit Set/Rest) mode is used to program individual bits of PORT C only.
49. How many pins of the 8255 can be used as the I/O ports?
There are 3 ports available in the 8255 so 24 pins are available for the I/O ports pins.
50. If we want to resize a 1024 x 768 image to one that is 640 pixels wide with the same aspect ratio,
what would be the height of the resized image?
 $\text{height} = 640 \times 768 / 1024 = 480$.
51. What is an image's aspect ratio?
The ratio of its width to its height, measured in unit length or number of pixels
52. If we use direct coding of RGB values with 2 bits per primary color, how many possible colors do we have for each pixel?
 $2^2 \times 2^2 \times 2^2 = 4 \times 4 \times 4 = 64$.
53. If we use 2-byte pixel values in a 24-bit lookup table representation, how many bytes does the lookup table occupy?
 $2^{16} \times 24 / 8 = 65,536 \times 3 = 196,608$.
54. What is Arduino?
Arduino is an open-source hardware and software platform that is easy to use. Arduino boards are capable of reading analog or digital input signals from various sensors.
55. How many types of Arduino do we have?
There are 4 Arduino boards and 4 Arduino shields that fit on top of Arduino compatible boards to provide additional capability like connecting to the internet, motor controller, LCD screen controlling, etc..
56. What language is a typical Arduino code based on?
The Arduino code is basically a lightly modified version of the C++ programming language. It includes certain functions or modules that are specific to the development of the Arduino platform and was preinstalled in the language by the developers.
57. What language is the Arduino IDE built on?
The Arduino code is primarily a subset of the C++ programming language. However, the IDE (Integrated Development Environment) was written in the Java programming

language. An IDE is a software that provides primarily a graphical user interface for compiling and maintaining/editing code for a platform.

58. How many analog pins are used in Arduino Mega board?

It has lots of digital input/output pins, 14 can be used as PWM output 16 analog inputs, a USB connection, a power jack, and a reset button.

59. Arduino IDE consists of 2 functions. What are they?

Setup() is called once in the program when a sketch starts after power-up. It is used to initialize variables, input and output pin modes, and other libraries needed in the sketch. Loop() is used after setup() been called, function loop() is executed repeatedly in the main program. It controls the board until the board is powered off or is reset.

60. Arduino Codes are referred to as _____ in the Arduino IDE.

Any Arduino Code that is saved using the Arduino IDE, is referred to as a “sketch” by the Arduino IDE. These “sketches” are nothing but the code saved in a file with the extension name being “.ino”.

61. What is the default bootloader of the Arduino UNO?

The optiboot bootloader will take 512 bytes, leaving 32256 bytes for application code. Due to its small size larger up-loadable sketch size is achieved.

62. What is the use of the Arduino.h header file?

The Arduino.h file is not something that we encounter during the process of writing the code for the Arduino board because the preprocessor takes care of it before compilation. However, it is a very important file without which we cannot access the Arduino’s features like GPIO IO, Analog IO, etc.

63. What is the use of the Vin pin present on some Arduino Boards?

The Vin Pin can be used for accepting a 9V power supply which is usually given through the external Jack. It can also be used for getting a 9V output when the Arduino is powered through the Jack, thus acting as both a power source and a short from the power source both when needed.

64. What is the execution process of an Arduino code?

The code that is written is first done so with the help of the editor. Without the editor there will be no place where the programmer can write his or her code. After that the code is handed over to the preprocessor and finally the compiler which translates the code into the Arduino’s Assembly Instruction Set.

65. What is the microcontroller used in Arduino UNO?

ATmega328p is a microcontroller which is 32KB of flash ROM and 8-bit microcontroller.

66. What does p refer to in ATmega328p?

Picopower technology employs advanced features like multiple clock domains, DMA and event systems to minimize power consumption.

67. Which board is the first to use a microcontroller within the build USB?

The Leonardo is Arduino’s first development board to use one microcontroller with built-in USB. This means that it can be cheaper and simple, And also, code libraries are available which allow the board to emulate a computer keyboard etc.

68. Which Arduino Board contains an onboard joystick?

The Arduino Esplora comes with the Atmega32u4 and is based on the Leonardo Hardware. It comes preinstalled with a lot of things including a joystick, buttons, TFT

Connector for attaching to external LCD's, an RGB LED, etc. Due to its configuration and setup, it is perfect for beginners to use.

69. What is the function of the IOREF pin on the Arduino UNO?

The IOREF pin provides the operation voltage reference for the Arduino. Usually on most Arduino Boards that is 5V but on the Diecimila it is 3.3V however. But this pin should not be at any given time connected to a load as power source.

70. Which processor supports the Arduino Zero?

The Arduino Zero uses the Atmel Smart SAM D21 MCU featuring the 32-bit ARM Cortex M0+ CPU. It has a flash memory of 256K bytes and an SRAM of 32K bytes.

71. Which software is used to upload the Arduino Sketches to the board?

The AVR-GCC/G++ software is a compiler which can take C/C++ code as input and translate or compile it into Binary Code that follows the instruction set for the AVR Microcontrollers. After that process is done, the avrdude software has the job of uploading the hex file to the microcontroller, more specifically the AVR Microcontroller family.

72. What is the use for the 2 serial pins on the Arduino Diecimila?

There are two pins for sending and receiving the Serial TTL Signals. The TX (for transmission) and the RX (for receiving). In the Arduino Diecimila they are connected to the corresponding pins of the FTDI USB to Serial Chip.

73. Which Arduino Boards use the Atmega2560?

Both the Arduino Mega and the Arduino Mega ADK use the Atmega2560, however the only difference between the two boards is that the Arduino Mega ADK was specifically built with Android phones in mind and has a USB to host connector that can plug into your android smartphone.

74. What is the operating voltage of Atmega328?

The operating voltage of the Atmega328 is 1.8V to 5.5V. Provide any voltage lower than that and the chip will not function. But however, if any voltage higher than the rated voltage is provided there is a chance of damaging the IC physically. So, under no circumstance should this chip be provided a voltage more than the rated voltage.

75. Which Arduino Boards use the Atmega32U4?

The Arduino Leonardo uses the Atmega32U4. Some other boards that use this microcontroller include the Arduino Pro Micro, which is a smaller development board than the Arduino Leonardo.

76. Is the Arduino code an Object-Oriented programming language or a Procedural programming language?

Since the Arduino Programming Language is a subset of the C++ Programming Language, it supports the Object-Oriented Programming approach much like C++.

77. What is the difference between an IDE and a compiler?

According to the primary definitions of a compiler and an IDE, the job of debugging and executing a piece of code falls on the compiler, while the job of the IDE is to provide an easy to use environment for writing the code in the first place.

78. What will be the output of the following Arduino code?

```

#define X 10;
void setup(){
    X=0;
    Serial.begin(9600);
    Serial.print(X);
}
void loop(){
    //Do nothing...
}

```

In the above code we have attempted to modify the value of X which is initialized using the preprocessing directive “define” which makes its value unchangeable throughout the entire code. Thus, the compiler will throw an error.

79. Where does the Arduino IDE search if it needs to find out the Name of a type of Arduino Board?

The different types of Arduino Boards which are also commonly referred to as “Variants” are present in the boards.txt file. This is a particularly important file because it contains the different configuration information for different Arduino Boards like which compiler toolchain to execute on a particular variant.

80. How many times does the setup() function run on every startup of the Arduino System?

The setup() function is used predominantly to configure the pins, variables, Serial data, etc. and is executed only once throughout the entire cycle of the program. However other than the above-mentioned uses, it can also be used to execute technically all aspects of an Arduino program, but since it only runs one time, it’s not very useful for anything other than configuring.

81. What is the output of “pin1” if “pin2” is sent “1011” where 1 is 5V and 0 is 0V?

```

1. int pin1 = 12;
2. int pin2 = 11;
3. void setup() {
4.     pinMode(pin1, OUTPUT);
5.     pinMode(pin2, INPUT);
6.     Serial.begin(9600);
7. }
8. void loop() {
9.     if(digitalRead(pin2)==1) {
10.         digitalWrite(pin1,LOW);
11.     }
12.     else if(digitalRead(pin2)==0)
13.         digitalWrite(pin1,HIGH);
14.     }
15. }

```

Here when pin2 receives a digital 1 we set pin1 as “LOW” or a digital 0, and conversely when pin2 receives a digital 0 we set pin1 as “HIGH” or a digital 1. Therefore, effectively generating the inverse of the wave that we received as input.

82. Which Arduino Board does the SigFox Library work with?

The SigFox Library allows the use of the SigFox transceiver with the Arduino MKRFOX1200 Board. It uses UNB (Ultra Narrow Band) to interconnect remote devices. It's a low energy device which falls in the category of LPWAN (Low Powered Wide Area Network).

83. What will be the output of the following Arduino code?

```
void main() {  
    int a = 0;  
    double d = 10.21;  
    printf("%lu", sizeof(a + d));  
}  
  
void loop() {}
```

The sizes of int and double are 4 and 8 respectively, a is an integer variable whereas d is a double variable. The final result will be a double in this case in order to keep the precision. Hence the output of the code is 8 bytes.

84. What is the use of the ESP8266 WiFi Module?

The ESP8266 WiFi Module is basically a network provider module which helps your micro-controller to connect to any WiFi network in the vicinity.

85. Why is the Arduino Mega more a viable solution when dealing with complex projects, than the Arduino UNO?

The Arduino UNO has 14 pins for IO of which 6 support PWM, while the Arduino Mega has 54 pins for IO of which 14 support PWM. Moreover, the Arduino Mega uses the Atmega2560 microcontroller which has an SRAM of 4K bytes and a flash memory of 256K bytes, whereas the Arduino UNO which is supported by the Atmega328P microcontroller has an SRAM of only 2K bytes and a flash memory of a mere 32K bytes.

86. The Atmega168 is an _____ bit chip.

The Atmega168 is capable of processing 8 bits of data at a time and hence it has an 8-bit architecture. Other similar 8-bit processors include the Intel 8080 Processor which ran on a similar 8-bit architecture.

87. Why does every compilation of a source code in Arduino check for previous compilations?

During the compilation process, the Arduino compiler before compilation searches for previously generated ".o" files in the directory, to make the compilation process faster in some cases. This reduces the redundancy of compiling the same parts of the code repeatedly.

88. What type of signal does the analogWrite() function output?

The Pulse Width Modulated Signal is the one which is transmitted by the analogWrite() function. This is done in an attempt to replicate the analog functionality to control certain sensors and actuators.

89. What is the objective of the code given below if it is executed on the Arduino Uno?

```

1. #include<EEPROM.h>
2. int pin=13;
3. void setup() {
4.     pinMode(pin,OUTPUT);
5.     Serial.begin(9600);
6. }
7. void loop() {
8.     for(int i=0;i<EEPROM.length()
9.         EEPROM.write(i, 1);
10.    }
11.    digitalWrite(pin,HIGH);
12.    exit(0);
13. }

```

The EEPROM.write() function is used to write data to the EEPROM directly. This function takes in 2 arguments; the address to which the data is to be written, and the data which is to be written to the EEPROM in the specified address. The program fills the EEPROM with 1's and then turns on the onboard LED of the Arduino Uno.

90. What is the use of the SD.h Library in Arduino?

The SD.h Library in Arduino allows for communications between the Arduino Board and an SD Card Module for storage and retrieval of data. It is dependent on the sdfatlib library which works on FAT16 and FAT32 filesystems.

91. What is the use of the Ethernet library?

The Ethernet Library is used to connect the Arduino to the internet via an Ethernet connection. This library is designed to be used with any Arduino Ethernet Shields or any other W5100, W5200, etc. based devices.

92. What is the purpose of the following Arduino code?

```

1. void setup() {
2.     Serial.begin(9600);
3. }
4. void setup() {
5.     Serial.write(40);
6. }

```

The above code uses the serial pins on the Arduino Board. The data is sent in the form of a byte whose value is 40. This data is processed into a data packet where the additional sections are added to number 40 and then send using UART.

93. Which chipset is the LCD library for Arduino based on?

The LCD Library in Arduino allows for controlling LCD Displays with the Arduino. This is based on the Hitachi HD44780 Chipset which is a dot matrix LCD (Liquid Crystal Display) Driver. It can be configured and used with 4-bit and 8-bit microcontrollers.

94. What is the resolution of the micros() function on the Arduino Nano?

The micros() function gives the up-time of the Arduino program in microseconds. This function has a resolution, i.e. the intervals in which it gives an output. This resolution depends on the Frequency of the specific Arduino Board that its running on. On the Arduino Duemilanove it's resolution is 4 Microseconds.

95. What is the use of the Interrupt Service Routine in an Arduino?

The interrupt service routine can be used to automate a number of functions in an arduino. In a situation where the entire focus of the code is bound to one function, the ISR can act as a pause for the arduino to stop the current function temporarily and divert memory to another function. That way, the entire memory of the arduino is used up in one function.

96. What is the output of the program given below if a voltage of 5V is supplied to the pin corresponding to the A0 pin on an Arduino UNO?

```
void setup() {  
    Serial.begin(9600);  
    pinMode(A0, INPUT);  
}  
void loop() {  
    int s = analogRead(A0);  
    Serial.println(s);  
}
```

The Arduino UNO's analog pins map the value of the sensed voltage to an internal numbering scale which makes it easier for the programmer to work since it is more difficult to analog voltage levels in programming practice. Thus, for a value of 5V, which reaches the maximum limit, we get a value of 1024.

97. What mode should we put the Arduino pin to, in order for object detection to work with the Ultrasonic Sensor?

The TRIGGER pin is to be kept HIGH for a period of 10 microseconds meanwhile the ECHO pin is HIGH for the time period it takes for the sent Ultrasonic wave to return to the sensor. Thus, the entire detection is digital.

98. What will happen if we supply a voltage of 25V to the Vcc of the Nokia 5110 GDM?

The Nokia 5110 Graphical Display Modules are mostly built to work on a voltage range of approximately 3.3V to 5V. Any voltage lower than that and the sensor will not be able to power on, but however any voltage significantly above that and the sensor may suffer permanent damage.

99. Which port on any Arduino board should be connected to the Key Port on the Adafruit FONA 3G+GPS Breakout?

The Adafruit FONA 3G+GPS Breakout's Key pin should be always connected to the Gnd Pin on any Arduino Board. This pin provides the function of the power on/off indicator. When you would like to shutdown the board, connect the pin to the Gnd of the Arduino Board for about 3 to 5 seconds. However for prolonged use usage of the Adafruit FONA 3G+GPS Breakout one needs to permanently connect this pin to the Gnd of the Arduino.

100. What is the way of throwing an error using preprocessing directives to the Arduino Compiler and forcing it to stop compilation?

The #error directive first throws an error to the Arduino IDE which is then displayed and then the compilation process is stopped prematurely. This is used to enable the user to stop the compilation process programmatically if some criteria or condition is not met before he or she can allow the program to start running.